Replication Report

Mode: full · Generated 2026-08-24 22:12

30%
score

Not reproduced

3/32 claims reproduced **12** partial **8** did not reproduce

2 fixes applied

Claims at a glance

Flags

- 1 not_attempted claim(s) excluded from the denominator as run/tooling-limited (blocked_infra / no_evidence), not counted against the paper.
- 2 claim(s) marked out_of_scope (planner did not target them) — excluded from the score, not counted as failures. Score is over the 32 attempted/blocked claim(s).

Limitations & code quality

2 fixes were needed to run the provided code (1 minor, 1 major, 0 critical). These describe the code, not the replication.

#	Fix	Severity
1	Added FRED API request throttling (0.9s minimum interval) and 429-aware backoff to avoid rate limits.	minor
2		major

#	Fix	Severity
	Replaced a set_index('year').loc lookup on a duplicated-year frame with a precomputed national ln-employment series indexed by year, fixing a regression size mismatch.	

Details

ID	Tier	Type	Status	How graded	Why
C1	headline	scalar	**match**	agent	The run produced results/ persistence_convergence.json with emp_persistence_slope=0.6976716605 and emp_persistence_r2=0.7542403243498 over emp_persistence_n=51 units (the 5 states + DC), from step 3 (src/ persistence_convergence....
C2	supporting	scalar	**match**	agent	The run re-ran the Figure 1 employment persistence regression (1970-90 growth 1950-70 growth) dropping Arizona, Flori and Nevada. results/ persistence_convergence.json reports emp_persistence_r2_ex_AZ_FL_NV = 0.6101290902950444, and...
C3	supporting	qualitative	**partial**	agent	Evidence is /workspace/output/replicatio codebase/results/adf_tests.json. The thre states the claim names (MA, SD, WY) all the unit root at 5% and all have negative lagged-level coefficients, matching the p substantive fin...
C4	headline	table	**partial**	agent	The run produced the Table 1 impulse responses in results/table1_irfs.json (the planned output/tables/table1.csv was no written, but the impulse_responses block the gradeable evidence). Grading per ce row/column label against pa...
C5	supporting	scalar	**no match**	agent	The run produced per-state long-run employment responses in results/ table1_irfs.json ('individual_long_run'), computed as the plateau 1/(1-sum_phi) AR(4) in relative-employment change. O four named outliers, only Wyoming repr

ID	Tier	Type	Status	How graded	Why
C6	supporting	scalar	**not attempted**	agent	The claim targets the exact regression o 1985 relative state unemployment rate 1975 rate (Figure 3; slope 0.03, t 0.2, R 0.00). This specific regression could NOT produced by the run: the pipeline reconstructed state une...
C7	supporting	scalar	**not attempted**	agent	The C7 regression requires the relative s unemployment rate at a 1970 endpoint, reconstructed state unemployment (LAU basis) is only available from 1976 onwar pipeline therefore explicitly recorded the '1990_on_1970' pair ...
C8	headline	scalar	**partial**	agent	Step 6 estimated the pooled AR(2) for re unemployment and produced the implie impulse response in results/table1_irfs.js impulse_responses.unemployment = {y 1.075, year2: 0.800, year3: 0.478, year4 0.230, year5: 0.077, ...
C9	supporting	qualitative	**match**	agent	results/adf_tests.json (wages block) repo n_states=51, n_negative_coef=48, n_reject_5pct=3 (reject_states=['NE','WI','WY'], crit_5pct=-2.94), which I independently recomputed from the 51 per_state entrie adf_tests.py confirms th...
C10	headline	scalar	**partial**	agent	The claim has two halves. (1) The perso income-per-capita convergence regressi (avg annual pcinc growth 1948-90 on log income) WAS reproduced in results/ persistence_convergence.json: pcinc_convergence_slope=-0.01197 (pa -0.0...
C11	supporting	scalar	**not attempted**	agent	C11 concerns the persistence of average MANUFACTURING wage growth: the cros correlation of 1950-70 mfg wage growth 1970-90 mfg wage growth (paper value The replication step 3 that targets C11 explicitly did NOT reconst...
C12	supporting	scalar	**no match**	agent	The run produced Table 2 in results/ table2_and_regionyear.json, but every q

ID	Tier	Type	Status	How graded	Why
					is far outside tolerance. Produced avg_a = 0.969 vs paper 0.66 (~47% off); n_states_beta_sig_diff_1 = 50 vs paper 1 per-state betas: AK 2.27 vs -1....
C13	supporting	scalar	**no match**	agent	Step 4 ran successfully (replication_log.j exit_code 0) and wrote results/ table2_and_regionyear.json, whose region_year block gives census_division_share=0.6419 (64%) an census_region_share=0.5704 (57%). The paper reports about 38% ...
C14	supporting	scalar	**no match**	agent	The run produced the regression underly Figure 6 in results/ unemployment_cross_section.json under avg_urate_vs_empgrowth: slope = -33.3 -1.71, r2 = 0.056, n = 51, over window '1976-1990'. The paper reports slope -0. with t-sta...
C15	headline	scalar	**no match**	agent	The run produced the trivariate VAR imp responses in results/trivariate_var.json. T reported year-1 responses to the -1% employment shock are year1.unemployment_response = 0.001994520587944889 and year1.participation_response = -0....
C16	headline	scalar	**partial**	agent	The trivariate VAR (step 8) produced res trivariate_var.json and figures/ fig7_trivariate_irf.png from the 1978-199 LAUS window (n_obs=663). The employ IRF builds to a peak magnitude of -0.021 (-2.1%) at year 3 (year 4 = -0.020...
C17	headline	scalar	**no match**	agent	Step 8 produced the year-1 per-worker decomposition in results/trivariate_var.js under 'per_worker' (basis: 'shares of the employment response'): unemployment=0.1152, participation=0 implied_net_migration=0.6007, derived
C18	supporting	scalar	**no match**	agent	Step 8 produced results/trivariate_var.js with okun.all_states = {coef: 18.270, t: 14.687, n: 714} and okun.farm_states = {coef: 14.667, t: 4.424, n: 126}. The pa

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					pooled Okun coefficient is ~-0.32 (t=17. the farm-state coeff...
C19	supporting	scalar	**no match**	agent	The claim concerns Figure 9's regressior average employment growth on relative MANUFACTURING wages, 1950-90 (slope -0.68, t=0.6). The run could not reconstr this: manufacturing production-worker A (SIC basis, 1947-1990) is flagge...
C20	supporting	scalar	**not attempted**	agent	Claim C20 requires the regression under Figure 10 -- average manufacturing wag average state unemployment rate (1970 expecting a positive relationship with t~ and R^2~0.16. The manufacturing production-worker wage seri...
C21	supporting	scalar	**partial**	agent	Step 9 (src/bivariate_irfs.py) produced re bivariate_irfs.json and figures/ fig11_emp_wage.png, so Figure 11 was generated and is gradeable. The EMPLO dynamics match the paper well: from a initial shock (year 0 = -0.01) em...
C22	supporting	scalar	**out of scope**	deterministic	No plan step targeted this claim; treated out_of_scope (excluded from the score, failure): The industry/worker-characteris controlled CPS wage series is an author- constructed set of annual state earnings regression residuals ...
C23	supporting	scalar	**partial**	agent	Step 9 (bivariate employment-personal- income system, Figure 13/C23) produced results/bivariate_irfs.json with emp_inco giving the per-capita personal income response to a -1% employment shock ov years 0-20. The path is normalized...
C24	supporting	scalar	**partial**	agent	The run produced results/bivariate_irfs.j with suppress_feedback = {employment_plateau_with_feedback: -0.011717, employment_plateau_without_feedback -0.014579}, and the emp_wage IRF at y gives the initial employment change of -

ID	Tier	Type	Status	How graded	Why
C25	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the pla marked it BLOCKED (counts as a reproducibility gap): Requires NAR Home Yearbook median metro house prices (39 MSAs, 1969-90) that Poterba supplied privately to the authors; this proprie...
C26	supporting	scalar	**out of scope**	deterministic	No plan step targeted this claim; treated out_of_scope (excluded from the score, failure): Reconstructable in principle fro metro CPI-U (23 MSAs) plus a matching employment panel, but it is a specialized supporting appe...
C27	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the pla marked it BLOCKED (counts as a reproducibility gap): The real-consumpti wage calculation is derived from the hou price (C25) and CPI (C26) impulse respo it inherits the blocked pro...
C28	supporting	figure	**not attempted**	deterministic	No plan step targeted this claim; the pla marked it BLOCKED (counts as a reproducibility gap): Requires the Davis-Loungani-Mahidhara defense (military p contract) series and the Bartik two-digit industry-mix instrument, both priv...
C29	supporting	scalar	**partial**	agent	The run executed step 10 (src/massachusetts_case.py, exit_code 0) an produced results/massachusetts.json wit growth_1980_85 = 1.985% and gap_vs_us_1980_85 = 0.495 pp. The MA 1980-85 annual growth of ~1.98% matc the paper's 2.0% almo...
C30	supporting	scalar	**partial**	agent	Only the unemployment sub-component this five-part claim was reconstructable run. results/massachusetts.json reports unemp_rel_1979 = -0.35 (paper: equal t national, i.e. 0 — close) and unemp_rel_ = -3.0917, i.e. 3.09 pp bel...
C31	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the pla marked it BLOCKED (counts as a reproducibility gap): Boston relative hou

ID	Tier	Type	Status	How graded	Why
					prices come from the same proprietary I metro median-price series as C25; not reconstructable from reachable pub...
C32	supporting	scalar	**partial**	agent	results/massachusetts.json (produced by 10, src/massachusetts_case.py) reports reconstructed MA employment panel: avg_growth_1985_91 = -0.617, gap_vs_us_1985_91 = -2.382 (i.e. ~2.38 below national), growth_1990 = -4.269,
C33	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the pla marked it BLOCKED (counts as a reproducibility gap): The Massachusetts bust simulation predicts nominal-wage a Boston housing-price growth using the estimated wage/housing system, whic...
C34	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the pla marked it BLOCKED (counts as a reproducibility gap): Describes propertie (state-income shares, forecast-error vola of the Davis-Loungani-Mahidhara militar prime-contract-award series...
C35	supporting	figure	**partial**	agent	The exact expected file output/figures/ figure2.pdf was not produced, but a rele figure figures/fig2_cumulative_rel_emp.p was, titled 'Fig 2: cumulative relative employment by region group', plotting cumulative relative log-employme...

Python 3.12.8 · GPU NVIDIA RTX A6000 (x4, 49140 MiB each) · pandas 3.0.5, numpy 2.5.2, statsmodels 0.14.6, scipy 1.18.1, matplotlib installed

Step	What it did	Result	Time
1	Create working copy, venv, install deps, make output dirs.	**ok**	40s
2	Reconstruct 51-unit state panel 1947-1990 from FRED/BLS.	**ok**	113s
3	Persistence & convergence regressions (Figs 1,2,4,9).	**ok**	4s
4	State-national employment elasticities (Table 2) + region-year variance decomposition.	**ok**	3s
5		**ok**	3s

Step	What it did	Result	Time
	ADF unit-root tests on relative employment (trend, 4 lags) and relative wages (no trend, pcinc proxy), 1952-1990.		
6	Pooled univariate models & implied impulse responses (Table 1) + per-state long-run employment responses.	**ok**	3s
7	Unemployment persistence & cross-state correlations (Figs 3,6,10), LAUS 1976-1990.	**ok**	3s
8	Trivariate VAR (emp, unemployment, participation) Fig 7 + Okun coefficient (C18).	**ok**	3s
9	Bivariate emp-wage & emp-income systems (Figs 11,13,14).	**ok**	3s
10	Massachusetts case study (descriptive reconstructable parts).	**ok**	6s

The full machine-readable record is on disk next to this report:

- `verify/verdicts.json` — every claim verdict, with the grader's cited rationale.
- `verify/replication_score.json` — the score and tier breakdown.
- `replication/replication_log.json` — step-by-step execution, with the fixes applied.
- `replication/codebase.diff` — every change made to the provided code.
- `replication/codebase/` — the patched code that was actually run.